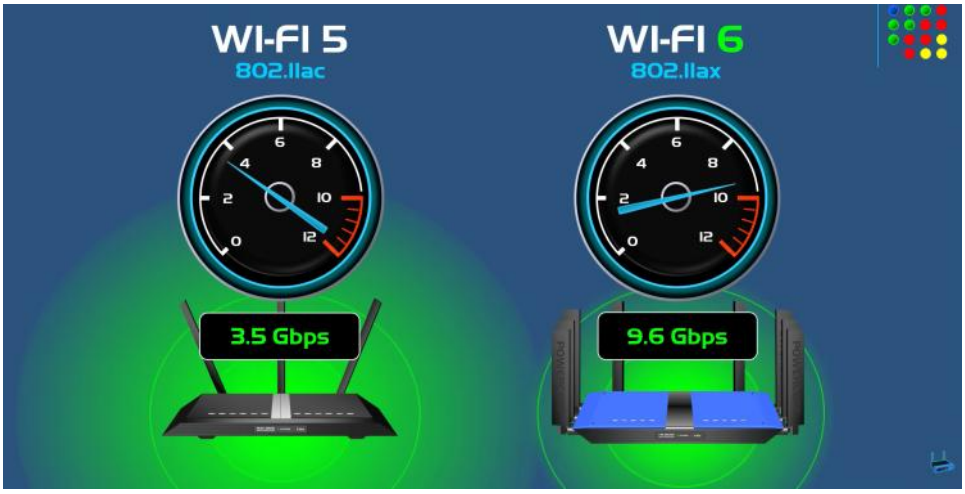
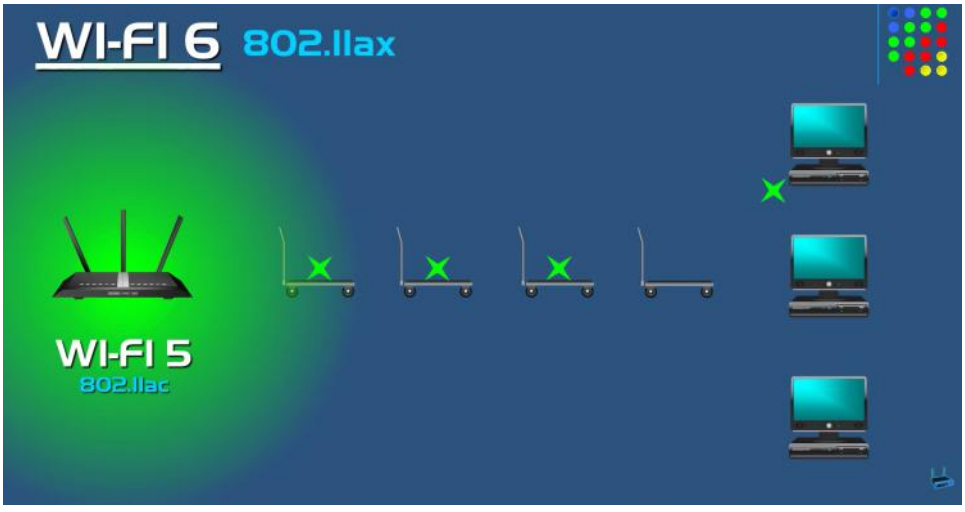
Wi-Fi 6

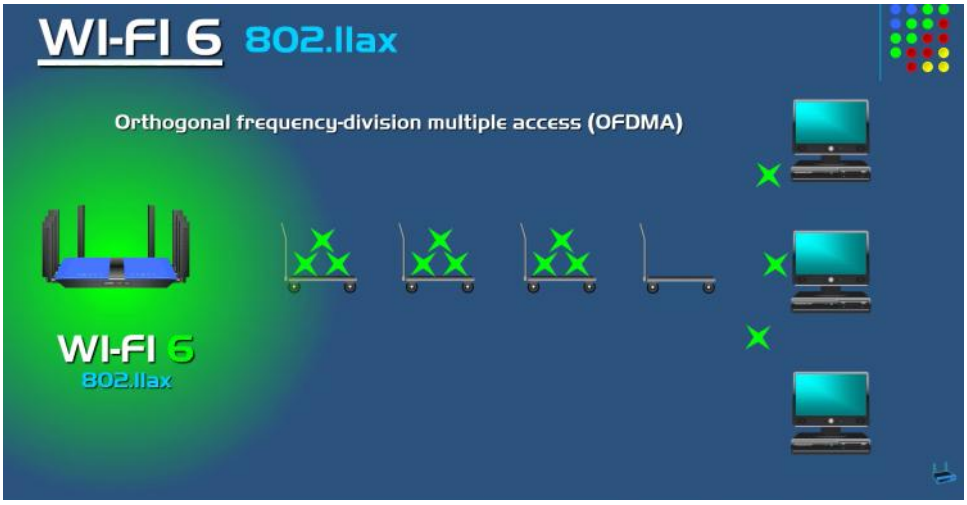
Friday, March 27, 2026 6:49 AM

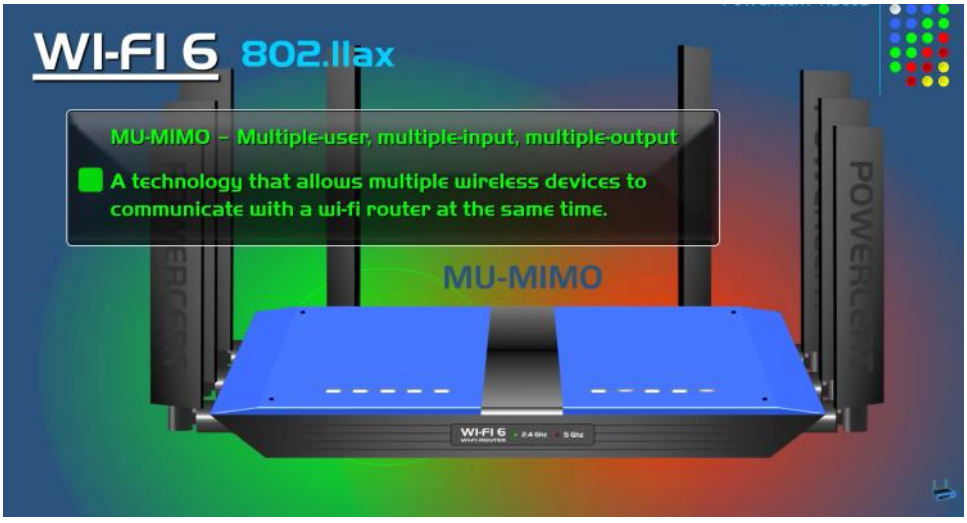
Wi Fi 6 also known as 802.11ax is a newer wireless standard released in 2019 that improves network performance especially when many devices are connected



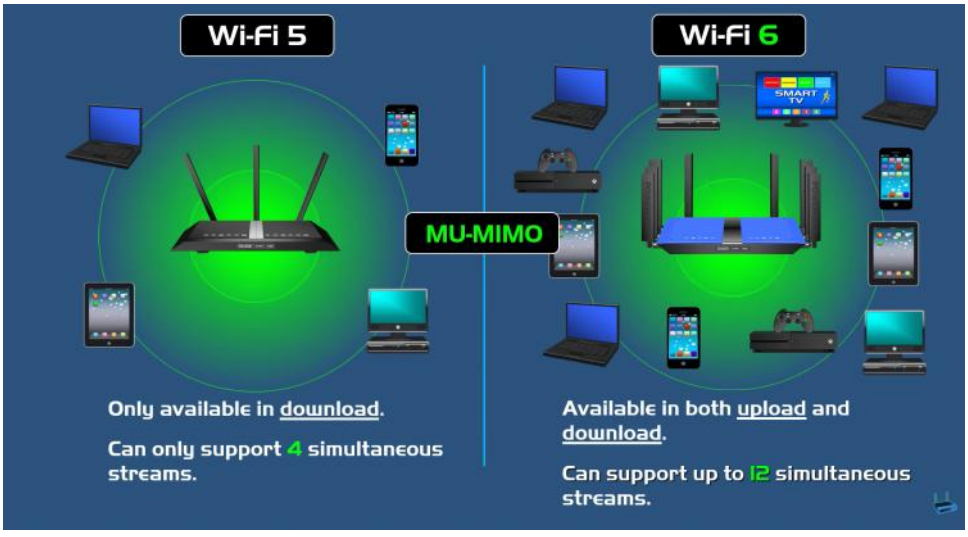
It offers higher total speed up to 9.6 Gbps but the main goal is better efficiency not just speed

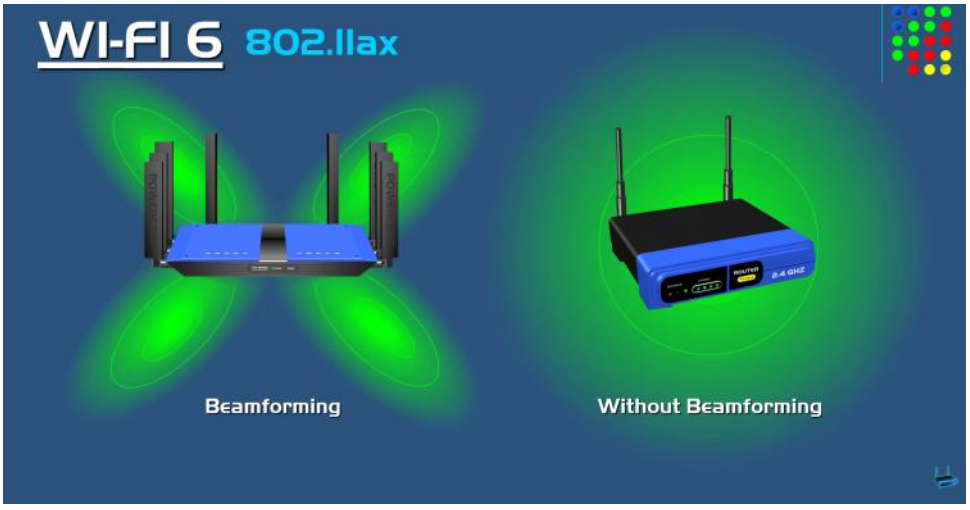



WI-FI 6 802.11ax
Orthogonal frequency-division multiple access (OFDMA)
WI-FI 6
802.11ax


WI-FI 6 802.11ax
MU-MIMO – Multiple-user, multiple-input, multiple-output
A technology that allows multiple wireless devices to communicate with a wi-fi router at the same time.
MU-MIMO

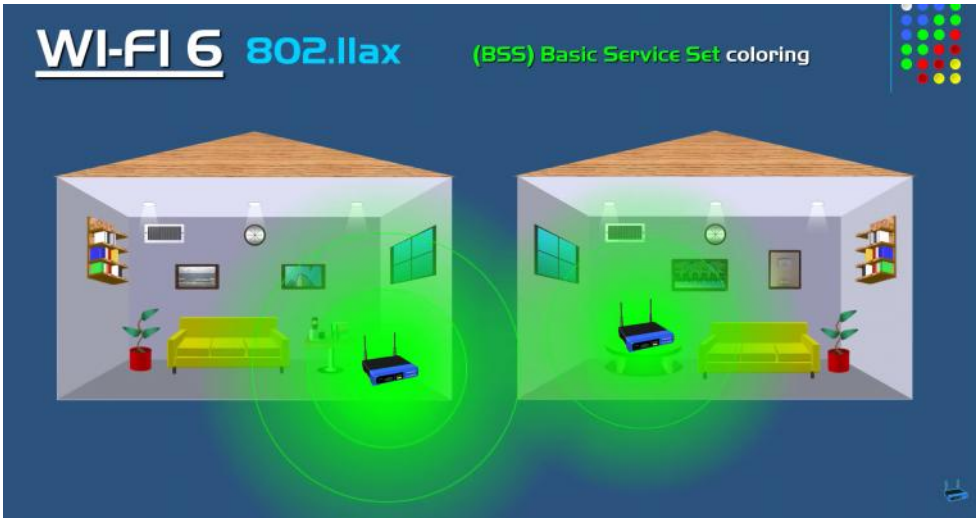
WI-FI 6 802.11ax
(Prior to WI-FI 5)
SU-MIMO - Single-user, Multiple-Input, Multiple-output.
Communicates with multiple devices but only one at a time.

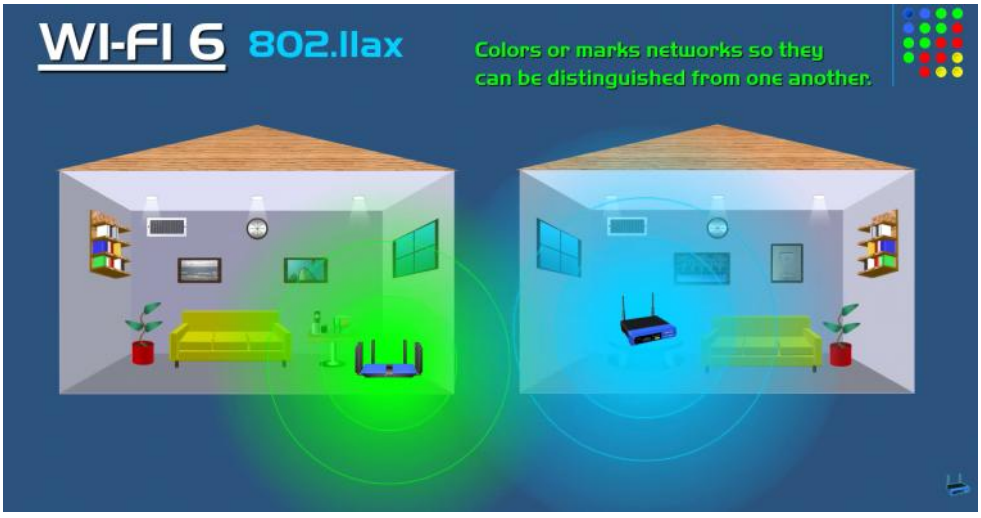




Key technologies include OFDMA which allows multiple devices to receive data at the same time reducing delay and MU MIMO which enables simultaneous communication with multiple devices for both upload and download

Beamforming is better as it will send signals to a particular way rather than getting the signals to all the ways





It also improves signal strength using beamforming reduces interference using BSS coloring and enhances security with WPA3

Wi Fi 6 can extend device battery life using Target Wake Time and supports more devices efficiently

Wi Fi 6E adds a new 6 GHz band which provides more channels and better performance in crowded environments

Overall Wi Fi 6 is best for homes or offices with many connected devices and gives a faster more stable and efficient wireless experience